Adventures in Reproducibility with Nix

Nigel Stanger

March 27, 2026

University of Otago
ŌTĀKOU WHAKAIHU WAKA
School of Computing
Te Kura Rorohiko

Section 1

The problem

Monday...

```
JAVA_HOME="/Applications/Apache NetBeans.app/Contents/Home"
cd /Users/stani07p/LocalDocuments/Development/INFORMS3/service; ../gradlew --configure-on-demand -x check run
Configuration on demand is an incubating feature.
> Task :service:compileJava UP-TO-DATE
> Task :service:processResources UP-TO-DATE
> Task :service:classes UP-TO-DATE

> Task :service:run FAILED
Exception in thread "main" io.jooby.exception.StartupException: Application startup resulted in exception
        at io.jooby.Jooby.start(Jooby.java:815)
        at io.jooby.Jooby.runApp(Jooby.java:1119)
        at io.jooby.Jooby.runApp(Jooby.java:1082)
        at web.Service.main(Service.java:47)
Caused by: java.lang.IllegalArgumentException: Unsupported class file major version 69
        at io.jooby.internal.$shaded.asm.ClassReader.<init>(ClassReader.java:200)
        at io.jooby.internal.$shaded.asm.ClassReader.<init>(ClassReader.java:180)
        at io.jooby.internal.$shaded.asm.ClassReader.<init>(ClassReader.java:166)
        at io.jooby.internal.RouteAnalyzer.returnType(RouteAnalyzer.java:44)
        at io.jooby.internal.RouterImpl.start(RouterImpl.java:555)
        at io.jooby.Jooby.start(Jooby.java:865)
        at io.jooby.Server$Base.fireStart(Server.java:93)
        at io.jooby.netty.Netty.start(Netty.java:97)
        at io.jooby.Jooby.start(Jooby.java:805)
        ... 3 more

FAILURE: Build failed with an exception.
```

Tuesday...

```
> source("~/LocalDocuments/Teaching/2025/INFO408/results/assessment_plots.R")
Error in library(ggplot2) : there is no package called ‘ggplot2’
```

assessment_plots.R

Source on Save

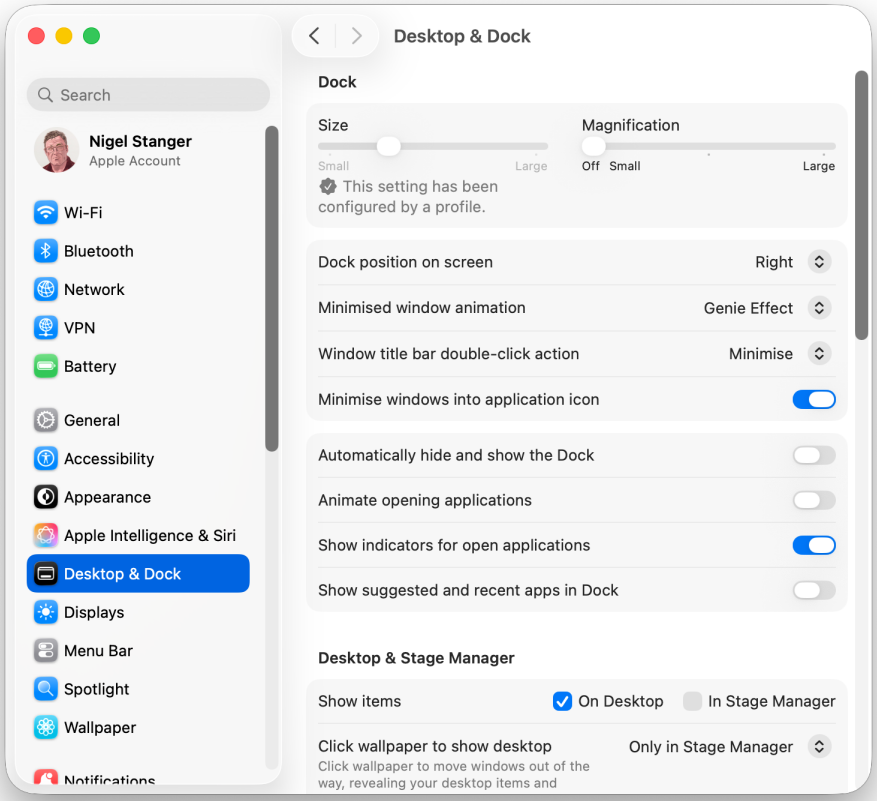
Packages ggplot2, readr, and scales others required but are not installed. Install Don't Show Again

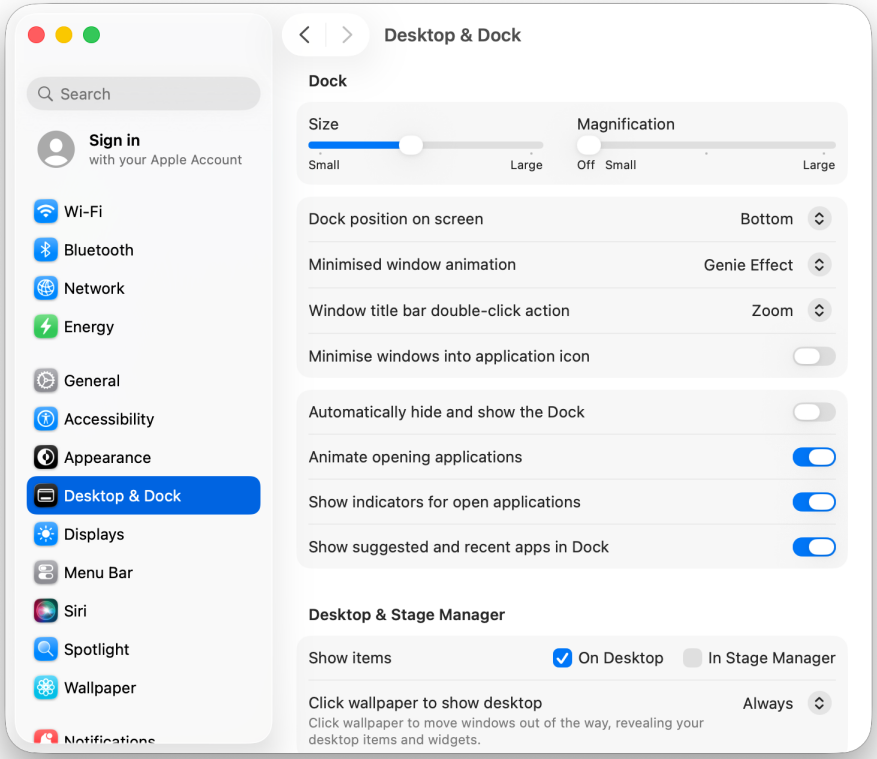
```
1 library(ggplot2)
2 library(scales)
3 library(readr)
4 library(tidyr)
5 library(tibble)
```

Wednesday...

```
›› python INFORMS/find_broken_names.py
Traceback (most recent call last):
  File "/Users/.../INFORMS/find_broken_names.py", line 6, in <module>
    import petl as etl
ModuleNotFoundError: No module named 'petl'
```

Meanwhile...


Desktop & Dock
Search
Nigel Stanger
Apple Account
Wi-Fi
Bluetooth
Network
VPN
Battery
General
Accessibility
Appearance
Apple Intelligence & Siri
Desktop & Dock
Displays
Menu Bar
Spotlight
Wallpaper
Notifications
Dock
Size
Small
Large
Magnification
Off
Small
Large
This setting has been configured by a profile.
Dock position on screen
Right
Minimised window animation
Genie Effect
Window title bar double-click action
Minimise
Minimise windows into application icon
Automatically hide and show the Dock
Animate opening applications
Show indicators for open applications
Show suggested and recent apps in Dock
Desktop & Stage Manager
Show items
On Desktop
In Stage Manager
Click wallpaper to show desktop
Only in Stage Manager
Click wallpaper to move windows out of the way, revealing your desktop items and


Desktop & Dock
Search
Sign in
with your Apple Account
Wi-Fi
Bluetooth
Network
Energy
General
Accessibility
Appearance
Desktop & Dock
Displays
Menu Bar
Siri
Spotlight
Wallpaper
Notifications
Dock
Size
Small
Large
Magnification
Off
Small
Large
Dock position on screen
Bottom
Minimised window animation
Genie Effect
Window title bar double-click action
Zoom
Minimise windows into application icon
Automatically hide and show the Dock
Animate opening applications
Show indicators for open applications
Show suggested and recent apps in Dock
Desktop & Stage Manager
Show items
On Desktop
In Stage Manager
Click wallpaper to show desktop
Always
Click wallpaper to move windows out of the way, revealing your desktop items and widgets.

Thank you

Questions?